

Notes: Stein (JF, 1997)

Internal Capital Markets and the Competition for Corporate Resources

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1 Big picture

Question. Why should several technologically distinct projects be placed inside one firm and financed by corporate headquarters rather than each being set up as a stand-alone company that raises funds externally? Conditional on having an internal capital market, how large and how focused should it be?

Setting. The paper studies a credit-constrained environment in which project managers privately observe project quality, derive private benefits from investment, and therefore overstate how much funding their projects deserve. Headquarters can monitor projects, raise funds from outside investors, and then redistribute those funds across divisions.

Core friction. External financiers know managers want too much capital because private benefits rise with resources under control. As a result, financiers ration credit and some good projects are underfunded.

Main conceptual contribution. Headquarters can create value through *winner-picking*: taking scarce funds away from relatively weak projects and reallocating them toward relatively strong projects. This is the distinctive function of an internal capital market.

Main baseline result. Headquarters may add value even when it does *not* relax total firm-wide credit constraints at all. With two projects and a fixed aggregate funding pool, headquarters still improves efficiency by reallocating the same total funds better than arm's length markets do.

Main size result. The optimal size of an internal capital market trades off two forces. More projects under one roof can ease credit constraints and improve funding capacity, but too many projects weaken headquarters' monitoring effectiveness.

Main scope result. A focused strategy can sometimes dominate unrelated diversification. The reason is subtle: internal capital markets care about *relative* ranking of projects. If focus makes headquarters' mistakes more correlated across related projects, relative ranking may actually improve even if absolute forecasting accuracy does not.

Main organizational result. Headquarters differs from a bank because it has control rights that allow it to enforce exclusivity and engage in "loser-sticking" - forcing a weak project to accept less capital than it could try to raise on its own. This authority is what makes winner-picking feasible.

Economic takeaway. Internal capital markets are valuable not because headquarters is benevolent, but because a broad-scope empire-builder can sometimes be less parochial than individual project managers. Scarce capital is allocated better when a strong central allocator can compare projects and shift resources across them.

2 Model environment and key objects

2.1 External financing benchmark: founder, manager, and outside investors

The primitive unit is a single project. A founder has the initial idea or patent but no wealth and no managerial ability. To implement the project, the founder must hire a professional manager and obtain financing from outside investors.

The objective is to maximize the founder's expected net proceeds from the patent, after payments to financiers and wages to the manager. For simplicity, the manager's reservation wage and the financiers' required rate of return are both normalized to zero.

Interpretation. The paper strips the problem down to the minimal ingredients needed to generate external financing frictions: somebody with a project, somebody with private information, and outside financiers who cannot directly observe project quality.

2.2 Technology and states

Investment can be either 1 or 2 units. The project's verifiable cash flow depends on the state of the world:

- In the bad state B , investing 1 yields Y_1 , and investing 2 yields Y_2 .
- In the good state G , productivity is scaled up by $\theta > 1$, so investing 1 yields θY_1 , and investing 2 yields θY_2 .

The paper assumes

$$1 < Y_1 < Y_2 < 2$$

and

$$\theta(Y_2 - Y_1) > 1.$$

The first inequality says that in the bad state the founder prefers investment 1 to investment 2. The second says that in the good state the second unit of investment is socially valuable, so the first-best calls for investment 2.

The ex ante probability of the good state is p ; the bad state occurs with probability $1 - p$.

Interpretation. The project has state-contingent optimal scale. The bad state wants one unit, the good state wants two. Any financing arrangement that cannot condition cleanly on the state will either underinvest in G or overinvest in B .

2.3 Manager private benefits and the source of credit rationing

The manager observes the state directly, but outside investors do not. The manager enjoys private benefits that are proportional to project output:

- in state B , private benefits are sY_1 if investment is 1 and sY_2 if investment is 2;
- in state G , private benefits are $s\theta Y_1$ if investment is 1 and $s\theta Y_2$ if investment is 2.

These private benefits are unverifiable and cannot be assigned by contract to outsiders.

Because the manager prefers more resources under his control, he has an incentive to exaggerate project quality and ask for too much funding. Outside investors understand this tendency and therefore become skeptical.

Interpretation. The manager does not merely want positive NPV investment. He wants scale. This wedge between social returns and private control benefits is what creates the credit-rationing problem.

2.4 Why the external market may fund only one unit

If no revelation scheme is used, outside investors cannot make investment contingent on the state. They must either always supply 1 or always supply 2.

If they always supply 1, expected net output is positive:

$$(p\theta + (1 - p))Y_1 - 1 > 0.$$

Hence one unit can always be financed - for example, by a debt claim promising repayment 1.

If they always supply 2, expected net output is

$$(p\theta + (1 - p))Y_2 - 2.$$

The paper assumes parameters such that this is less attractive than always supplying 1. In that region, the optimal unconstrained external-market policy is to fund exactly one unit no matter what the true state is.

Interpretation. Outside investors ration credit not because they hate risk, but because they cannot distinguish genuine good news from self-serving optimism. They solve the problem crudely by capping funding at 1.

2.5 Why revelation schemes are set aside

In principle, one could try to pay the manager to reveal the state truthfully. But truthful revelation is costly because in the bad state the manager must be compensated for giving up the extra private benefits associated with overinvestment.

Stein notes that the ex ante cost of eliciting truthful information is

$$(1 - p)s(Y_2 - Y_1),$$

while the ex ante benefit is

$$p(\theta(Y_2 - Y_1) - 1).$$

The paper focuses on the region where private benefits are large enough, or the good state is rare enough, that revelation schemes are not worthwhile.

Interpretation. The model is deliberately placed in the hard case: outside markets cannot cheaply bribe truth out of managers, so a purely arm's length financing arrangement is stuck with underinvestment in good states.

2.6 Headquarters as a distinct organizational agent

An internal capital market introduces a new player: corporate headquarters. Headquarters has four defining features:

1. it can monitor projects and learn their prospects;
2. it has no wealth of its own;
3. it has control rights that let it appropriate a fraction $\phi < 1$ of project private benefits;
4. it has the authority to redistribute resources across projects.

The control-rights assumption matters in two ways. First, by taking part of the manager's private benefits, headquarters weakens the manager's incentives. Stein captures this reduced effort in a simple reduced form: whenever headquarters is in charge, verifiable project cash flows are multiplied by a factor $k < 1$.

Thus, for example, if state B occurs and the project receives investment 1 under headquarters, verifiable output becomes kY_1 , headquarters gets private benefit ϕskY_1 , and the manager gets $(1 - \phi)skY_1$.

Second, headquarters can reallocate resources across projects. If it raises funds against project i 's patent, it may still send those funds to project j .

Interpretation. Headquarters is not just a monitor. It is an owner-controller with the authority to reassign capital. That authority is the heart of the internal capital market.

2.7 Why control rights create exclusivity

The key organizational difference between headquarters and a nonowner intermediary is exclusivity. Headquarters can prevent a weak project from seceding and seeking its own financing elsewhere. A bank cannot.

Interpretation. Winner-picking always has a dark side: if one project wins, another loses. Loser-sticking is only feasible if headquarters can keep losers inside the system and deny them access to stand-alone financing.

3 Winner-picking in the baseline two-project model

3.1 Why headquarters is useless with only one project

With a single project, headquarters is strictly dominated. The reason is simple. Headquarters faces the same basic agency problem as a project manager: because its own private benefits also rise with investment, it cannot credibly tell the outside market that the state is bad and only one unit should be raised.

So with one project, headquarters cannot obtain more financing than the manager could obtain directly. At the same time, it imposes the cost $k < 1$ by diluting managerial effort.

Interpretation. Internal capital markets are not justified by inserting a magical monitor between a project and the market. They only become useful when headquarters spans *multiple* projects and can compare them.

3.2 Two independent projects under one roof

Now suppose headquarters oversees two independent projects, i and j , each with the same technology as above.

Assume:

- the two projects' states are independent;
- headquarters monitors both projects perfectly;
- outside investors provide a total of exactly 2 units of financing to headquarters - no more than the two projects could raise separately.

With total funding fixed at 2, headquarters can choose among allocations such as $(1, 1)$, $(2, 0)$, or $(0, 2)$.

Interpretation. This is the pure winner-picking case. Headquarters is not bringing in more

money. It is only deciding how to use the same money better.

3.3 External market benchmark

Under stand-alone external financing, each project always receives one unit. Expected total net output from the two projects is therefore

$$EM = 2[Y_1(p\theta + (1-p)) - 1]. \quad (1)$$

Interpretation. Equation (1) says the external market does the same thing to both projects regardless of their realized relative quality. It protects itself against overinvestment in bad states, but mechanically underinvests in good states.

3.4 Internal market benchmark

Under headquarters, output is multiplied by k , but headquarters can reallocate the two units toward the winner when the two projects are in different states. Expected total net output becomes

$$IM = 2(1-p)^2 k Y_1 + 2p^2 k \theta Y_1 + 2p(1-p) k \theta Y_2 - 2. \quad (2)$$

Each term has a clean interpretation:

- with probability $(1-p)^2$, both projects are bad, so the efficient use of the 2 units is still 1 and 1;
- with probability p^2 , both are good, but because only 2 units are available, headquarters again splits resources symmetrically as 1 and 1;
- with probability $2p(1-p)$, one project is good and the other is bad, so headquarters sends 2 units to the good project and 0 to the bad one.

Interpretation. The mixed-state case is where the internal capital market earns its keep. The same total funding is now concentrated on the winner.

3.5 The key efficiency condition

For reallocation to be genuinely valuable, the second unit invested in a good project must beat the first unit invested in a bad one:

$$\theta(Y_2 - Y_1) > Y_1.$$

Interpretation. This condition says the marginal return to concentrating capital on a good project exceeds the marginal return to spreading capital evenly across all projects. It is the pure winner-picking condition.

3.6 Why headquarters has the right incentives to pick winners

In the stylized baseline, headquarters' private benefits are proportional to project cash flows. Therefore, when deciding which project gets the scarce second unit, headquarters maximizes its own private benefits by maximizing project cash flow.

Interpretation. Headquarters is still self-interested and still wants scale on average, but conditional on having a fixed pool of funds, broader control makes it less parochial than any individual

project manager. The strong project gains because headquarters benefits from total firm output, not just from one division's empire.

3.7 Investment interdependence inside the firm

A crucial implication follows immediately: the investment of project i depends not only on its own prospects, but also on the prospects of project j .

If project j 's appeal rises, project i can lose capital even when project i remains positive NPV at the margin.

Interpretation. Internal capital markets create cross-project spillovers by construction. This is not a bug of the theory; it is the mechanism. Scarce resources are assigned comparatively, not project by project in isolation.

3.8 Why headquarters is different from a bank

A bank can monitor, but it cannot generally force a borrower to accept less financing than it could seek on its own. Headquarters can.

Interpretation. The bank's comparative advantage is delegated monitoring for fund-raising. Headquarters' comparative advantage is control-based reallocation. The paper is about the second margin.

4 Optimal size and scope of internal capital markets

4.1 Size: why larger internal markets can relax credit constraints

Once headquarters oversees many independent projects, there is a diversification-like force that can loosen total financing constraints. If headquarters controls a large portfolio, it can credibly use a larger total funding pool because some projects will be good and others bad, allowing capital to be shifted toward high-return uses.

In Stein's extreme illustration, if headquarters oversees 1000 independent projects, the law of large numbers implies that roughly $1000p$ projects will be in state G and roughly $1000(1 - p)$ will be in state B . In that case, $1000(1 + p)$ units of financing can nearly implement the first-best allocation: 2 units to each good project and 1 unit to each bad project.

Interpretation. More projects can make headquarters more valuable because capital can be packed more tightly into good uses.

4.2 Why the model needs a force limiting size

If that were the only effect, the model would predict an arbitrarily large firm. Stein therefore adds a second force: headquarters' monitoring ability deteriorates as it oversees more projects.

Let $M(n)$ be the probability that headquarters successfully monitors and learns the true state of each of n projects. $M(n)$ is decreasing in n . In Stein's simple formulation, monitoring is all-or-nothing:

- with probability $M(n)$, headquarters learns all project states;
- with probability $1 - M(n)$, headquarters learns nothing.

If total funding is F , expected per-project profit is

$$\pi(n, F) = M(n)\pi_M(n, F) + (1 - M(n))\pi_N(n, F), \quad (3)$$

where π_M is per-project profit when monitoring succeeds and π_N is per-project profit when it fails. The organization chooses $F^*(n)$ for each n , and then chooses the n that maximizes $\pi(n, F^*(n))$.

Interpretation. Size is pinned down by a classic tradeoff: more projects loosen aggregate financing constraints but weaken information quality.

4.3 Example 1: better monitoring can support a larger internal market

Stein's first numerical example sets

$$Y_1 = 1.01, \quad Y_2 = 1.68, \quad \theta = 2, \quad p = 0.4, \quad k = 1.$$

For these parameters:

- $F^*(2) = 2$ for any $M(2)$;
- $F^*(3) = 4$ if $M(3) > 0.538$, and $F^*(3) = 3$ otherwise.

With relatively weak monitoring, say $M(2) = 0.6$ and $M(3) = 0.5$, neither 2 nor 3 projects relax credit constraints, so the 2-project internal market is better because it monitors more accurately:

$$\pi(2) = 0.461, \quad \pi(3) = 0.454.$$

If monitoring improves to $M(2) = 0.9$ and $M(3) = 0.8$, a 3-project internal market can now raise more capital, and this benefit slightly dominates the loss in monitoring accuracy:

$$\pi(2) = 0.485, \quad \pi(3) = 0.487.$$

Interpretation. Better monitoring can push the firm toward a larger internal capital market because the extra project helps ease credit constraints.

4.4 Example 2: the optimal size can move nonmonotonically

The second example sets

$$Y_1 = 1.1, \quad Y_2 = 1.9, \quad \theta = 1.5, \quad p = 0.4, \quad k = 1.$$

Now:

- $F^*(2) = 3$ if $M(2) > 0.833$;
- $F^*(3) = 4$ if $M(3) > 0.361$.

With weak monitoring, $M(2) = 0.4$ and $M(3) = 0.3$, neither size eases constraints, so 2 projects are better.

With intermediate monitoring, $M(2) = 0.8$ and $M(3) = 0.6$, only the 3-project structure eases constraints, so 3 projects become better:

$$\pi(2) = 0.339, \quad \pi(3) = 0.343.$$

With very strong monitoring, $M(2) = 1$ and $M(3) = 0.65$, even the 2-project structure can now ease constraints, and the smaller organization again becomes optimal:

$$\pi(2) = 0.348, \quad \pi(3) = 0.346.$$

Interpretation. The optimal size need not increase monotonically with better monitoring. Once smaller firms can already raise enough, the extra complexity of additional projects is no longer worth it.

4.5 Scope: diversification versus focus when signals are perfect

If headquarters observes project outcomes perfectly, diversification appears to dominate focus. The extreme case is perfect focus with perfectly correlated project outcomes. Then one project is never better than the other, so winner-picking disappears.

Interpretation. If focus merely means “all projects move together,” it weakens the case for an internal capital market because there are fewer genuine winners and losers to sort across.

4.6 Scope with noisy signals and correlated errors

Stein argues that this first diversification result is too shallow, because focus may not primarily affect *outcomes*. It may affect headquarters’ *forecasting errors*.

To capture this, he keeps actual project outcomes independent but assumes headquarters receives noisy high/low signals about each project:

$$\mathbb{P}(H \mid G) = \mathbb{P}(L \mid B) = q, \quad \frac{1}{2} < q < 1.$$

He then introduces a focus parameter $a \in [0, 1]$ that makes *false* signals more likely to be correlated across projects. For example, if project j gets a false low signal when its true state is good, project i is more likely to get a false low signal as well:

$$\mathbb{P}(L_i \mid G_i, G_j, L_j) = (1 - q)(1 + a) > (1 - q) = \mathbb{P}(L_i \mid G_i). \quad (4)$$

Analogously, if project j gets a false high signal when its true state is bad,

$$\mathbb{P}(H_i \mid B_i, B_j, H_j) = (1 - q)(1 + a) > (1 - q) = \mathbb{P}(H_i \mid B_i). \quad (5)$$

Stein also imposes simplifying restrictions such as

$$\mathbb{P}(L_i \mid B_i, G_j, L_j) = q, \quad (6)$$

and

$$\mathbb{P}(H_i \mid G_i, B_j, H_j) = q. \quad (7)$$

Interpretation. Focus means headquarters tends to make the *same* mistake across related projects. Relative ranking can therefore remain useful even if absolute evaluation is noisy.

4.7 The proposition on focus

Using the 16-state appendix table, Stein proves the following proposition:

Proposition. The value created by the internal capital market is increasing in the focus parameter a . This positive effect is stronger when q is smaller, that is, when signals are less accurate.

Interpretation. When project-evaluation errors are highly correlated, headquarters is less likely to receive misleadingly different signals about equally attractive projects. Common errors often do little damage to relative ranking; asymmetric errors do.

4.8 Why focus can help even when absolute forecasts are noisy

The intuition is easiest to see by contrasting two kinds of mistakes:

- **Common-error states** such as *BBHH* or *GGLL*: headquarters is wrong about both projects in the same direction.
- **Asymmetric-error states** such as *BBHL* or *GGLH*: the two projects are actually similar, but headquarters receives different signals and reallocates capital unnecessarily.

Raising a shifts probability mass toward the first type and away from the second. That helps because internal capital markets care mostly about *relative* ranking, not absolute levels. If both projects are misjudged together, the ranking is often unaffected.

Interpretation. Focus can improve winner-picking because it synchronizes mistakes. Correlated mistakes are less harmful than uncorrelated mistakes when the job is to decide which project is better.

4.9 Pulling the size and scope results together

The paper's final message on organization is not "bigger is always better" or "focus is always better." Instead:

- size is chosen by balancing funding capacity against monitoring quality;
- scope is chosen by balancing the diversification of outcomes against the correlation of evaluation errors.

This also creates a wedge between founder-optimal and manager-chosen organizations. Managers who care about private benefits may prefer larger, more diversified organizations than founders or shareholders would.

Interpretation. The model is normative at its core, but it also hints at why observed diversification can be excessive from the standpoint of firm value.

5 Discussion, key assumptions, and tables/figures

5.1 What assumptions make the results work

The paper's results rely on five especially important assumptions:

1. **External markets are meaningfully credit constrained.** If outside investors can cheaply

elicit truth or perfectly audit managers, the need for an internal capital market shrinks sharply.

2. **Private benefits are positively related to project value.** Headquarters overinvests on average, but conditional on fixed resources it still tends to prefer stronger projects.
3. **Headquarters has real control rights.** Without authority to deny stand-alone financing to weak projects, loser-sticking and hence winner-picking collapse.
4. **Monitoring does not scale costlessly.** This is what prevents the model from predicting an arbitrarily large firm.
5. **Focus correlates forecast errors more than true outcomes.** This is what makes focus beneficial in Section III.B.

Interpretation. The theory is not “all conglomerates are good.” It is a conditional statement about settings where capital is scarce, control rights are strong, and headquarters can genuinely compare projects.

5.2 When internal capital markets should matter most

Stein argues that internal capital markets should be most useful when outside finance is especially underdeveloped. In his language, that means environments where accounting, auditing, or legal protection are weak and private-benefit extraction is severe, so the parameter s is effectively high.

Interpretation. When external markets are bad at screening and disciplining managers, the value of reallocating scarce capital inside the firm rises. As external markets improve over time, the comparative advantage of internal capital markets should weaken.

5.3 Headquarters versus a bank lender

Diamond-style banking models emphasize delegated monitoring that allows the intermediary to bring more or cheaper finance to borrowers. Stein’s headquarters is different.

- A bank creates value primarily by easing credit constraints.
- Headquarters can create value even with a fixed funding pool.
- A bank ideally holds many unrelated loans.
- Headquarters may optimally oversee a small and even focused set of projects.

Interpretation. The model is specifically about *internal* capital allocation because the authority to reassign corporate resources is central. Monitoring alone is not enough.

5.4 Table 1: external versus internal allocation in the two-project benchmark

Table 1: How the two-project internal capital market reallocates a fixed funding pool

State pair	External market allocation	Internal market allocation	Economic point
BB	$(1, 1)$	$(1, 1)$	No winner exists, so reallocating does not help.
GG	$(1, 1)$	$(1, 1)$	With only 2 total units, headquarters cannot fully fund both good projects.
GB or BG	$(1, 1)$	$(2, 0)$ or $(0, 2)$	Headquarters sends the scarce second unit to the winner, which is where winner-picking creates value.

Read.

- Only the mixed-state cases distinguish internal and external capital markets.
- The internal market improves allocation without increasing total funds.
- The mechanism is inherently comparative: one project's gain is another's loss.

Economic message. This table is the paper in miniature. Headquarters matters only because it can compare projects and reassign fixed resources.

5.5 Table 2: the two numerical examples for optimal size

Table 2: How monitoring quality changes the preferred number of projects

Example	Parameter core	Funding thresholds	Main lesson
1	$Y_1 = 1.01$, $Y_2 = 1.68$, $\theta = 2$, $p = 0.4$	$F^*(2) = 2$, $F^*(3) = 4$ if $M(3) > 0.538$	Better monitoring can make a larger 3-project internal market optimal because only then do credit constraints relax.
2	$Y_1 = 1.1$, $Y_2 = 1.9$, $\theta = 1.5$, $p = 0.4$	$F^*(2) = 3$ if $M(2) > 0.833$, $F^*(3) = 4$ if $M(3) > 0.361$	The optimal size can be nonmonotonic: first 2, then 3, then back to 2 as monitoring improves.

Read.

- Size is not pinned down by one force alone.
- More projects help by easing total funding constraints.
- Fewer projects help by preserving monitoring accuracy.

Economic message. Internal capital markets should not be thought of as mechanically scale-loving. The optimal size is an organizational tradeoff, not an accounting identity.

5.6 Figure 1: appendix table for focus and correlated forecast errors

Appendix				
The Value of an Internal Capital Market as a Function of the Degree of Focus				
	Outcome/Signal Configuration	Probability	Payoff: External Market	Payoff: Internal Market
1.	GGHH	$p^2(q^2 + \alpha(1 - q)^2)$	$2\theta y_1$	$2\theta y_1$
2.	GBHH	$p(1 - p)q(1 - q)$	$y_1(1 + \theta)$	$y_1(1 + \theta)$
3.	BGHH	$(1 - p)p(1 - q)q$	$y_1(1 + \theta)$	$y_1(1 + \theta)$
4.	BBHH	$(1 - p)^2(1 - q)^2(1 + \alpha)$	$2y_1$	$2y_1$
5.	GGHL	$p^2(q(1 - q) - \alpha(1 - q)^2)$	$2\theta y_1$	θy_2
6.	GBHL	$p(1 - p)q^2$	$y_1(1 + \theta)$	θy_2
7.	BGHL	$(1 - p)p(1 - q)^2$	$y_1(1 + \theta)$	y_2
8.	BBHL	$(1 - p)^2(q(1 - q) - \alpha(1 - q)^2)$	$2y_1$	y_2
9.	GGLH	$p^2(q(1 - q) - \alpha(1 - q)^2)$	$2\theta y_1$	θy_2
10.	GBLH	$p(1 - p)(1 - q)^2$	$y_1(1 + \theta)$	y_2
11.	BGLH	$(1 - p)pq^2$	$y_1(1 + \theta)$	θy_2
12.	BBLH	$(1 - p)^2(q(1 - q) - \alpha(1 - q)^2)$	$2y_1$	y_2
13.	GGLL	$p^2(1 - q)^2(1 + \alpha)$	$2\theta y_1$	$2\theta y_1$
14.	GBLL	$p(1 - p)(1 - q)q$	$y_1(1 + \theta)$	$y_1(1 + \theta)$
15.	BGLL	$(1 - p)pq(1 - q)$	$y_1(1 + \theta)$	$y_1(1 + \theta)$
16.	BLL	$(1 - p)^2(q^2 + \alpha(1 - q)^2)$	$2y_1$	$2y_1$

Figure 1: Appendix table from the paper: outcome/signal configurations under external and internal markets

Read.

- The 16 rows enumerate all combinations of true outcomes and headquarters' signals.
- Rows such as *BBHH* and *GGLL* are *common-error* cases: both projects are misread in the same direction.
- Rows such as *BBHL*, *BBLH*, *GGHL*, and *GGLH* are the more damaging *asymmetric-error* cases in which headquarters mistakenly ranks equally attractive projects differently.

Economic message. Focus improves the internal capital market when it shifts probability mass toward common mistakes and away from asymmetric ranking mistakes. Relative comparison is what matters.

6 Short summary

- External finance is rationed because project managers privately observe quality and like larger empires.
- A one-project headquarters is useless: it faces the same financing friction and also weakens managerial effort.
- A multi-project headquarters can create value through winner-picking, even if it raises no more money than stand-alone projects could.

- The critical organizational ingredient is control: headquarters can reallocate funds and force weak projects to accept less capital.
- Optimal size balances looser aggregate credit constraints against worse monitoring as more projects come under one roof.
- Diversification is not always best. Focus can help if it makes forecast errors correlated across related projects and therefore preserves relative ranking.
- Internal capital markets should matter most when external capital markets are weak and agency problems are severe.

7 Conclusion

7.1 Core conclusion

Stein's central claim is that an internal capital market is justified when headquarters can allocate scarce capital across competing projects better than arm's length markets can. This need not require headquarters to raise more total funds. Reallocation alone can be enough.

7.2 Economic intuition

The same empire-building motive that makes managers overinvest can be partly harnessed at a higher level of the hierarchy. A project manager is too attached to his own project to surrender capital voluntarily. Headquarters, because it derives private benefits from a broader portfolio, can be more willing to move funds away from relative losers and toward relative winners.

7.3 What the paper does and does not deliver

The paper gives a sharp rationale for internal capital markets, but it does so in a stylized reduced-form framework. The monitoring technology, the effort-dilution cost k , and the error-correlation structure are imposed rather than derived from a deeper microfoundation. So the paper is best read as clarifying a mechanism, not as fully characterizing every real conglomerate.

7.4 Takeaway

Internal capital markets are most valuable when capital is scarce, comparisons across projects matter, and headquarters has enough control and information to engage in disciplined winner-picking. Bigger and more diversified is not automatically better; sometimes the best internal capital market is relatively small and relatively focused.